

Enhancing fingerprint template security using elliptic curve cryptography and fuzzy vault encoding

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ABSTRACT

Fingerprint authentication is widely employed for identity verification due to its reliance on unique biometric traits, such as ridge patterns and minutiae points. Despite its advantages, traditional fingerprint systems remain susceptible to spoofing and privacy breaches, raising significant security concerns. To mitigate these risks, cancelable biometrics have emerged as a viable solution by irreversibly transforming biometric features, thereby preventing the reconstruction of the original fingerprint even in the event of template compromise. This paper proposes a novel approach for generating cancelable biometric templates from fingerprint images by transforming minutiae points and securely mapping them onto elliptic curves. Authentication is performed entirely within the transformed domain using a fuzzy vault scheme, thus preserving template security throughout the matching process. The efficacy of the suggested method is demonstrated by experimental evaluations on the FVC2002 and FVC2004 datasets, which reveal EER values (in %) of 0.0, 0.02, and 5.0 for DB1, DB2, and DB3 of FVC2002 and 2.4 for DB1 of FVC2004, respectively. The security and comparative analysis prove the effectiveness of the proposed method over current methods.

1. Introduction

Modern security technologies have made great strides in biometric authentication systems (BAS), which provide a reliable substitute for traditional knowledge-based (passwords, for example) or token-based (ID cards, for example) authentication techniques. BAS offers improved identity verification with increased resistance against prevalent threats like credential theft and impersonation by utilizing distinctive physiological or behavioral traits, such as fingerprints, iris patterns, palm-prints, or facial features. At the core of these systems lies the biometric template: a compact and discriminative digital representation of an individual's biometric traits, which serves as the basis for comparison during the authentication process (see Fig. 1).

Despite their advantages, the deployment of BAS introduces serious security and privacy concerns, particularly regarding the protection of stored biometric templates. Various attack strategies have shown that biometric templates, if compromised, can be reverse-engineered to reconstruct original biometric traits, leading to unauthorized access and irreversible loss of biometric confidentiality [1,2]. Among the most prominent attacks are template inversion attacks, where adversaries attempt to reconstruct biometric traits from stored templates;

cross-matching attacks, where templates across different databases are correlated to track individuals without their consent; and spoofing attacks, where fabricated biometric samples are used to impersonate legitimate users. These vulnerabilities highlight the pressing need for robust, privacy-preserving protection schemes for biometric data.

Cancelable biometric techniques have become a viable way to solve privacy and security issues in biometric systems. In cancelable biometrics, the original biometric data is subjected to a non-invertible transformation, generating an application-specific pseudo-identity for storage and matching. As defined by the ISO/IEC 24745:2011 standard, two essential properties must be upheld: irreversibility and cancelability [3]. Even if the transformation parameters are revealed, irreversibility guarantees that it is computationally impossible to recreate the original biometric data from the altered template. Cancelability allows for the generation of multiple, distinct templates from the same biometric source, facilitating revocation and preventing cross-matching across different applications. Preserving high recognition accuracy within the transformed domain is crucial to maintaining system usability while ensuring robust template protection.

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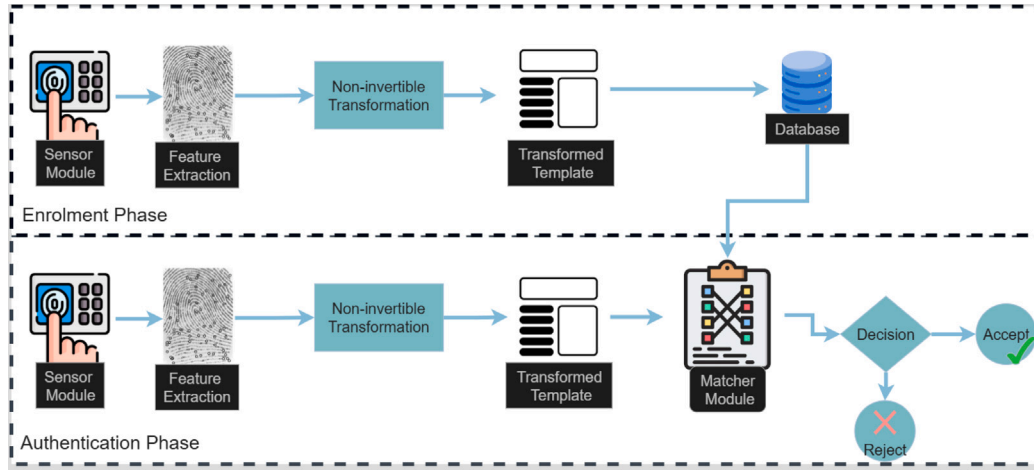


Fig. 1. Biometric authentication system.

In this paper, we propose a novel method for generating secure cancelable fingerprint templates by integrating elliptic curve cryptographic transformations of minutiae features with a Fuzzy Vault-based authentication framework. The use of elliptic curve cryptography (ECC) offers strong security guarantees with efficient computation, while the Fuzzy Vault scheme provides robust authentication even with intra-class variability and prevents unauthorized access. The combination of ECC and fuzzy vault systems provides a powerful and efficient solution for biometric authentication security. ECC provides robust cryptographic security with smaller key sizes than RSA, allowing for faster computations and reduced storage requirements, ideal for mobile and IoT devices. The fuzzy vault approach improves biometric robustness by allowing approximate matching while safeguarding templates with chaff points, limiting inversion attacks, and enabling template revocation. Combining these technologies results in a tiered security system, with ECC protecting cryptographic keys and the fuzzy vault protecting biometric templates. This technique protects privacy, assures scalability, and allows for secure regeneration if hacked. This framework, designed for real-world applications, facilitates distributed systems by balancing efficiency and security in resource-constrained contexts while protecting biometric data from potential breaches. Its versatility makes it a good fit for modern authentication methods. The proposed approach aims to enhance template security, ensure revocability, and maintain high authentication accuracy, thus addressing critical challenges in the design of next-generation biometric systems.

The remainder of this paper is structured as follows: Section 2 presents a review of related work. Section 3 outlines the essential background concepts necessary for understanding the proposed approach. Section 4 describes the proposed methodology in detail. Section 5 discusses the experimental setup, results, and performance evaluation. Section 6 offers a comprehensive security analysis of the system. Finally, Section 7 concludes the paper and suggests potential directions for future research.

2. Related work

Fingerprint-based biometric systems offer significant advantages over traditional authentication methods, such as passwords and tokens. However, one of the major concerns with these systems is the potential for identity theft, as fingerprint data cannot be changed once compromised. Furthermore, research has shown that fingerprint images can be reconstructed if minutiae points are exposed. In order to address these

challenges and enhance the security of fingerprint templates, several techniques have been proposed.

Ratha et al. [4] proposed several constructs for implementing cancellable biometrics in fingerprint recognition, focusing on feature domain transformations. Their work demonstrated the possibility of generating a large number of cancellable fingerprint templates with minimal loss in recognition accuracy, thereby increasing the security of biometric data.

Gao et al. [5] facial features extracted by deep neural networks contain personally identifiable information that can even be recreated from facial photos. This calls for strong biometric template protection (BTP) systems that adhere to requirements including accuracy preservation, revocability, unlinkability, and non-invertibility. In order to produce stable and cancelable face templates, the authors present a novel approach that combines Simhash and Multiple Partial Walsh Transformations (MPWT).

Sandhya and Prasad [6] proposed a method for constructing cancelable fingerprint templates by integrating bit strings derived from both local and global attributes of minutiae points. In a related line of work, Ali et al. [7] introduced a rotation-invariant, non-invertible transformation technique, which was subsequently refined in their later study [8]. Furthermore, Ali et al. [9] presented the Polynomial Vault framework, which employs polynomial curves for secure key binding within biometric systems. Although this approach offers strong security guarantees, it was observed to perform poorly under same-key attack scenarios.

Jignesh et al. [10] presented an alignment-free fingerprint authentication technique that uses Elliptic Curve Signcryption to improve the system's computational efficiency and security by combining minutiae point descriptors with Scale-Invariant Feature Transform (SIFT) keypoint descriptors.

Champaneria et al. [11] presented a cancellable biometric authentication scheme based on geometric transformations. Their approach leverages non-reversible transformations to secure fingerprint templates, protecting user identities from unauthorized access. This study emphasizes the advantages of geometric transformations in making templates resistant to reconstruction attacks, thus ensuring privacy and security in biometric systems.

M.S.Khan et al. [12] discusses how important it is to improve security and privacy in biometric systems, especially for palm print templates. Because compromised biometric data is irreplaceable and current cancelable palm print templates are weak in terms of security

and recognition performance, there are serious privacy and security hazards.

Khan et al. [13] discuss the use of fingerprint recognition systems has sparked serious worries about the security and privacy of biometric data, especially in light of the potential for compromised fingerprint templates. Through the transformation of original biometric data into a new, unlikable form that may be regenerated or cancelled in the event of breach, cancelable biometric systems present a promising potential solution.

Song et al. [14] The usage of fingerprints and faces in biometric systems is common, but maintaining these distinct and reliable templates presents serious security and privacy problems because leaked templates may allow original photos to be recovered. A solution is provided by cancelable biometric techniques, which convert original data into unlinkable, revocable, and irreversible versions.

Zhou et al. [15] demonstrate a workable and efficient multi-biometric system that provides a solid solution for secure, comprehensive identity management by effectively striking a balance between strong privacy protection and high recognition performance, particularly satisfying the requirements of irreversibility, unlinkability, and revocability for cancelable biometric templates.

Sp et al. [16] introduced a modality-independent approach for generating cancelable biometric templates based on random feature vector transformations. The method involves computing distance vectors between multiple randomized transformations of the original biometric feature vector, thereby ensuring non-invertibility and eliminating the need to store any original biometric data.

The fuzzy commitment scheme, proposed by Juels and Wattenberg [17], integrates error-correcting codes with cryptographic primitives to enhance the security of biometric templates. In this scheme, the binary representation of a user's biometric data is combined with a codeword from an error correction code using a bitwise exclusive-OR (XOR) operation, and the resulting value is stored as the protected template. During authentication, a new biometric sample is similarly binarized and XORed with the stored data to recover the original codeword, provided the input is sufficiently close to the original. This enables both secure key retrieval and reliable identity verification, even in the presence of minor biometric variations.

The fuzzy vault scheme, introduced by Juels and Sudan [18], secures a secret key by binding it with a set of genuine biometric data points – such as fingerprint minutiae – augmented with a large number of randomly generated chaff points. These chaff points obscure the genuine data, rendering the vault unintelligible to unauthorized users. During authentication, if the presented biometric data sufficiently overlaps with the original feature set, the system can successfully reconstruct the key and verify the user's identity, thereby enabling secure and error-tolerant biometric authentication.

Nandakumar [19] proposed a fuzzy commitment-based cryptosystem in which a mathematical function representing minutiae information is used to compute the binary representation of fingerprint data. The approach involves quantizing the Fourier spectrum of this function to generate a robust binary string, which is subsequently secured using the fuzzy commitment scheme. To address alignment issues, high-curvature points are utilized for aligning the probe fingerprint, and the corresponding alignment data is stored as helper information. While this technique enhances matching accuracy, the storage of auxiliary data introduces potential vulnerabilities, as it may be exploited in certain attack scenarios.

Li et al. [20] introduced a method for deriving the binary representation of fingerprint features, which are subsequently protected using the fuzzy commitment scheme. The binary string is generated from minutiae triplets, where each triplet consists of three minutiae points that meet a specific threshold criterion, such as the area of the triangle formed by the points. While this approach offers strong security, it necessitates multiple fingerprint samples and involves a complex procedure for generating the binary template, which could

be considered a limitation in terms of both efficiency and practical implementation.

Joni Saputra et al. [21] proposed a distance-based fuzzy vault method to enhance the accuracy of biometric authentication systems. Their approach incorporates several modifications, including minutiae filtering and candidate point identification techniques, which eliminate the need for image alignment and rotation during both registration and authentication. Additionally, the distance-based method avoids the generation of auxiliary helper data, which could otherwise pose a risk of information leakage, thereby reducing the potential for exploits by impostors.

Vivek Singh et al. [22] proposed a method for filtering genuine vault points from a combined set of genuine and chaff points, thereby enhancing the security of stored fingerprint templates. This technique ensures that only genuine minutiae contribute to the key recovery process, reducing the risk of unauthorized access. Additionally, to address minutiae misalignment during verification, they introduced a Principal Component Analysis (PCA)-based alignment technique, which significantly improves the accuracy of the verification process.

In [23], Nandakumar introduced a fuzzy vault technique that utilizes transformed minutiae point information, enabling revocability of the biometric template. This transformation is password-based, and a Cyclic Redundancy Check (CRC) is employed to detect errors in the generated key during the decoding process in authentication. This approach enhances both the security and reliability of the system by ensuring key integrity.

Li and Hu [24] proposed an alignment-free technique for securing fingerprint templates using the fuzzy vault method. They utilized pair-polar structures of minutiae points as features and constructed separate vaults for each pair-polar structure. While this approach enhances security, the need to generate multiple vaults introduces additional complexity to the encoding and decoding processes.

Uludag and Jain [25] introduced a fuzzy vault method to secure fingerprint templates by replacing traditional error correction codes with a Cyclic Redundancy Check (CRC). During enrollment, minutiae locations are mapped onto a polynomial derived from a secret key, and polynomial reconstruction is achieved through Lagrange interpolation during authentication. In a subsequent extension of this work, Nandakumar et al. [26] enhanced the vault by incorporating orientation values alongside minutiae locations during encoding. Additionally, they implemented a coarse filtering technique to selectively retain genuine minutiae during the decoding process.

Rathgeb et al. [27] proposed a multi-biometric fuzzy vault technique that integrates fingerprint and facial recognition features. By combining multiple biometric modalities at the feature level before applying the fuzzy vault scheme, this approach enhances both the security and accuracy of the biometric system. The study demonstrates notable improvements in recognition accuracy and resistance to spoofing attacks by leveraging the complementary strengths of multiple biometric traits.

Tams et al. [28] proposed a method to defend against cross-matching attacks by arranging minutiae points and chaff points in a hexagonal grid structure. This arrangement complicates attempts to cross-match with vault points from other databases, thereby preventing the retrieval of the original minutiae points. Additionally, the security of the minutiae-based fuzzy vault against various attack vectors is further analyzed in their work.

Abdullahi et al. [29] conducted a comprehensive survey on attacks targeting Biometric Authentication Systems (BAS), examining various attack vectors, including inversion attacks, false acceptance attacks, and attacks involving record multiplicity. Their findings emphasize the urgent need for secure and cancellable biometric systems to safeguard the integrity and privacy of biometric data. Jin et al. [30] introduce a key binding scheme for minutiae-based fingerprint biometrics that

does not rely on error-correcting codes, and incorporates cancellable transformations. Additionally, the minutiae data is securely protected using a robust, non-invertible cancellable transform, which plays a vital role in mitigating various security and privacy threats.

Existing fuzzy vault-based fingerprint template protection schemes often lack robust resistance against cross-matching and correlation attacks. In addition, many such schemes rely on the storage of helper data, which can become a vulnerability if exploited by an adversary. In this work, we enhance the traditional fuzzy vault approach [18] by integrating quantized minutiae descriptors mapped onto an elliptic curve, thereby introducing a robust cryptographic layer. This integration significantly mitigates the risk of cross-matching and reconstruction attacks, ensuring that the original biometric template remains secure even in the event of compromise. Furthermore, the use of Elliptic Curve Cryptography (ECC) enables the generation of user-specific public keys for authentication, eliminating the need for conventional error correction codes such as CRC, which are commonly employed in prior fuzzy vault-based schemes.

3. Preliminary

3.1. Fuzzy vault scheme

Biometric cryptosystems must address the challenge of intra-user variability in biometric data. The *Fuzzy Vault* scheme, proposed by Juels and Sudan [18], is a widely adopted method for binding a cryptographic key with noisy biometric data in a secure manner. Unlike conventional cryptography, which expects exact keys, the fuzzy vault allows unlocking even when the input features are only approximately correct.

3.1.1. Overview

The fuzzy vault is a form of secure sketch that hides a polynomial corresponding to a secret (e.g., a cryptographic key) among a large set of random chaff points. Only someone with a sufficiently overlapping set of biometric features can recover enough genuine points to reconstruct the polynomial and retrieve the secret. It leverages the hardness of the polynomial reconstruction problem over noisy data, ensuring security even if the vault is exposed.

Formally, let S be a secret, and let $Q = \{q_1, q_2, \dots, q_t\}$ be the user's biometric features extracted from a fingerprint image. A polynomial $F(x)$ of degree k is generated from S , and each $q_i \in Q$ is evaluated under F to produce $(q_i, F(q_i))$. These are the genuine points. To hide these, a number of chaff points (x_j, y_j) are added such that $x_j \notin Q$ and $y_j \neq F(q_j)$.

The final vault V is the union of genuine and chaff points:

$$V = \{(q_i, F(q_i))\} \cup \{(x_j, y_j)\}$$

The vault is shuffled and stored. Only someone with a query set Q' that significantly overlaps with Q can reconstruct $F(x)$ and thus retrieve S (see Fig. 2).

3.1.2. Phases of the Fuzzy vault

The fuzzy vault scheme primarily consists of two phases: **Locking** and **Unlocking**. In the Locking phase, the vault is created by combining genuine biometric features with a secret. During the Unlocking phase, the goal is to retrieve the secret by attempting to open the vault using a query template.

3.1.2.1. Locking phase. In this phase, a secret is encoded into a polynomial, and a mixture of genuine and chaff points is created to construct the vault.

Algorithm 1 Fuzzy Vault Locking Algorithm

Input: Secret S , Genuine Feature Set $Q = \{q_1, q_2, \dots, q_t\}$, Number of chaff points n , Degree k

Output: Vault V

- 1: Encode S as coefficients of a polynomial $F(x)$ of degree k over a finite field $\mathbb{F}$
- 2: Initialize vault $V \leftarrow \emptyset$
- 3: **for all** $q_i \in Q$ **do**
- 4: Compute $y_i \leftarrow F(q_i)$
- 5: Add genuine point (q_i, y_i) to V
- 6: **end for**
- 7: **while** number of chaff points added $< n$ **do**
- 8: Randomly select $x_j \in \mathbb{F} \setminus G$
- 9: Select random $y_j \in \mathbb{F}$ such that $y_j \neq P(x_j)$
- 10: Add chaff point (x_j, y_j) to V
- 11: **end while**
- 12: Shuffle V randomly
- 13: **return** V

3.1.2.2. Unlocking phase. The unlocking phase attempts to reconstruct the original polynomial using the query biometric features. If enough correct matches are found, the polynomial can be interpolated and the secret is recovered.

Algorithm 2 Fuzzy Vault Unlocking with Lagrange Interpolation

Input: Vault $V = \{(x_i, y_i)\}$, Query points set $P' = \{p'_1, p'_2, \dots, p'_m\}$, Threshold k

Output: Recovered Secret S or Failure

- 1: Initialize candidate set $C \leftarrow \{(x, y) \in V \mid x \in P'\}$
- 2: **for all** subsets $C' \subseteq C$ of size $k+1$ **do**
- 3: Let $C' = \{(x_0, y_0), (x_1, y_1), \dots, (x_k, y_k)\}$
- 4: Construct polynomial $F'(x)$ using Lagrange interpolation:

$$F'(x) = \sum_{j=0}^k y_j \cdot \prod_{\substack{0 \leq m \leq k \\ m \neq j}} \frac{x - x_m}{x_j - x_m}$$

- 5: Extract coefficients $\{s_1, s_2, \dots, s_{k+1}\}$ from $F'(x)$
- 6: Concatenate to form $s = s_1 \parallel s_2 \parallel \dots \parallel s_{k+1}$
- 7: **if** s passes verification (e.g., reconstructs valid public key) **then**
- 8: **return** Success
- 9: **end if**
- 10: **end for**
- 11: **return** Failure

3.1.3. Security considerations

The security of the fuzzy vault lies in the difficulty of distinguishing genuine points from chaff points without substantial overlap between the feature sets. Without a query set that closely resembles the real biometric traits, an attacker cannot rebuild the polynomial, even if they manage to get into the vault. The presence of numerous decoy points makes it computationally infeasible for the attacker to derive the original data.

3.2. Elliptic Curve Cryptography (ECC)

Elliptic Curve Cryptography (ECC) is a public-key cryptosystem that provides high security with significantly smaller key sizes compared to traditional systems like RSA [31,32]. ECC operates over the algebraic structure of elliptic curves defined on finite fields. Its efficiency makes ECC particularly well-suited for resource-constrained environments, such as embedded systems and biometric cryptosystems, where computational power and memory are limited.

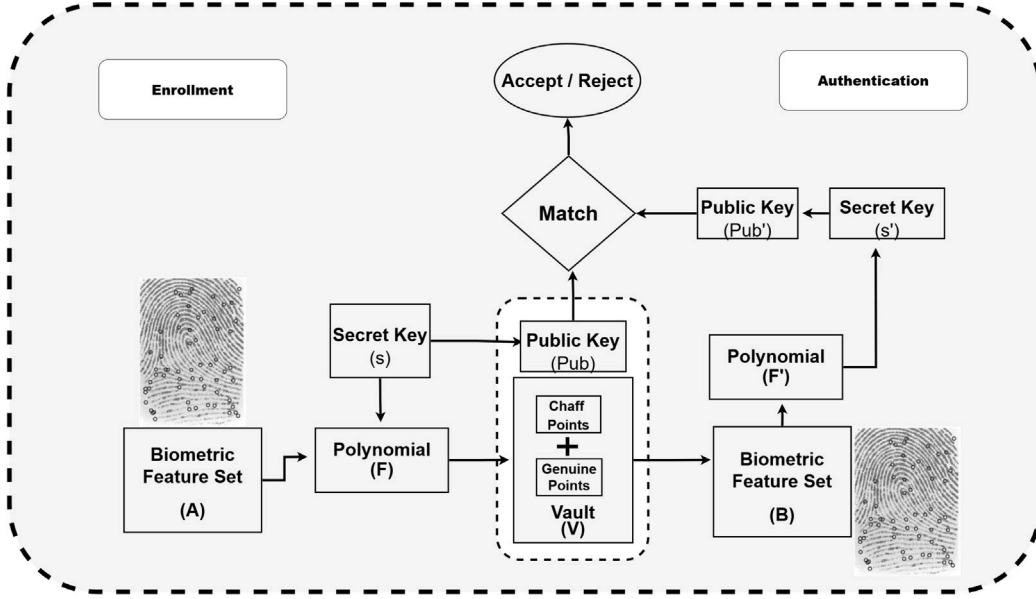


Fig. 2. Block diagram of fuzzy vault scheme.

An elliptic curve over a finite field $\mathbb{F}_p$ (where p is a prime) is defined by the Weierstrass equation:

$$y^2 = x^3 + ax + b \pmod{p}$$

where $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$ ensures that the curve has no singularities. The set of points (x, y) satisfying this equation, along with a special point at infinity $\mathcal{O}$, form an abelian group under a defined addition operation.

ECC derives its security from the Elliptic Curve Discrete Logarithm Problem (ECDLP): given points P and $Q = kP$ on the curve, it is computationally infeasible to determine k .

3.2.1. Mapping integers to curve points

In many ECC-based applications, including biometric systems, we need to bind integer values (e.g., hashed features, keys, etc.) to elliptic curve points. Since ECC operates on points, a deterministic mapping from an integer to a valid point on the curve is essential.

In elliptic curve-based cryptographic systems, it is often necessary to represent integer values as points on the curve. A simple and secure approach is to use scalar multiplication of the base point G with the integer x :

$$P = x \cdot G$$

Here, x is the input integer and G is the generator point defined on the elliptic curve. The result P is a valid point on the curve.

This representation is deterministic and produces a unique point for every integer modulo the order of G . Since scalar multiplication is a core operation in ECC and is computationally hard to reverse (due to the Elliptic Curve Discrete Logarithm Problem), it offers a secure and efficient way to bind an integer to a curve point [33].

More complex mappings like the Shallue–van de Woestijne (SWU) algorithm or Elligator offer uniform and privacy-preserving mappings but require more computational resources [34].

4. Proposed method

4.1. Image preprocessing and minutiae extraction

Fingerprint image preprocessing is a crucial step in ensuring reliable and accurate biometric feature extraction [1]. In our work, each fingerprint image is first resized to a standardized resolution of **500 × 500 pixels** and adjusted to a spatial resolution of **500 DPI**. This normalization ensures consistency across varying input sources and enhances the performance of the downstream minutiae extraction process.

To accurately align and extract fingerprint features, we utilize the **MegaMatcher SDK** [35], a commercial-grade biometric processing toolkit developed by Neurotechnology. The SDK's high-accuracy algorithms are employed to align the fingerprint images and extract the minutiae points. The resulting output is a structured set of minutiae descriptors, which includes ridge endings and ridge bifurcations. A ridge bifurcation minutiae is a position where a ridge divides into two pathways, and a ridge-ending minutiae is a point where a ridge ends. The i th minutiae point M_i of a fingerprint, which includes its x and y coordinates and its tracing direction θ , is defined by Eq. (1).

$$M_i = (x_i, y_i, \theta_i) \quad (1)$$

This preprocessing pipeline ensures that the input fingerprint templates are uniform and well-suited for further cryptographic transformation and matching.

4.2. Feature quantization and integer encoding

To convert fingerprint minutiae into a compact and privacy-preserving representation, we apply a quantization and encoding scheme. Each minutia point is represented by its spatial coordinates (x, y) and orientation θ . We normalize these values by dividing them with a quantization factor f , which reduces intra-class variations and increases robustness to noise:

$$x' = \left\lfloor \frac{x}{f} \right\rfloor, \quad y' = \left\lfloor \frac{y}{f} \right\rfloor, \quad \theta' = \left\lfloor \frac{\theta}{f} \right\rfloor$$

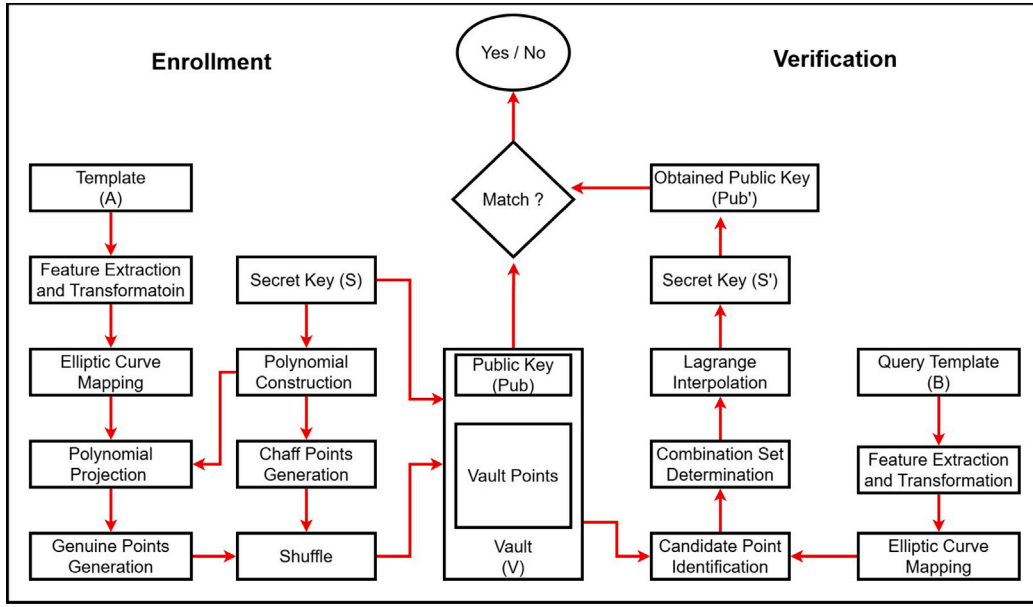


Fig. 3. Block diagram of proposed fuzzy vault based biometric authentication system.

The quantized values are then individually converted to their binary representations (6 bits each). These binary strings are concatenated to form a single bitstring (of 18 bits), which is finally converted to a decimal integer. This integer uniquely represents the minutia and is suitable for cryptographic binding in subsequent steps. This method ensures fixed-length encoding for each minutia point (see Fig. 3).

Algorithm 3 Quantize and Encode Minutiae as Integers

Input: Minutiae set $M = \{(x_i, y_i, \theta_i)\}$, quantization factor f

Output: Set of encoded integers Q

```

1:  $Q \leftarrow \emptyset$ 
2: for all  $(x, y, \theta) \in M$  do
3:    $x' \leftarrow \lfloor \frac{x}{f} \rfloor$ 
4:    $y' \leftarrow \lfloor \frac{y}{f} \rfloor$ 
5:    $\theta' \leftarrow \lfloor \frac{\theta}{f} \rfloor$ 
6:    $b_x \leftarrow \text{Binary}(x')$ 
7:    $b_y \leftarrow \text{Binary}(y')$ 
8:    $b_\theta \leftarrow \text{Binary}(\theta')$ 
9:    $b \leftarrow b_x \parallel b_y \parallel b_\theta$  ▷ Concatenate binary strings
10:   $q \leftarrow \text{Decimal}(b)$ 
11:   $Q \leftarrow Q \cup \{q\}$ 
12: end for
13: return  $Q$ 

```

4.3. Elliptic curve mapping

Upon converting each minutia into a quantized integer, we subsequently map this integer onto an elliptic curve through scalar multiplication. This process ensures that the integer is transformed into a valid point on the elliptic curve.

An elliptic curve of large prime field P (256 bits) is chosen. A Generator point G of the curve is determined. Let q_i be the encoded integer from the previous step. The corresponding point P on the curve is computed as:

$$P = q_i G$$

This operation binds the biometric data into the algebraic structure of the elliptic curve, leveraging the hardness of the Elliptic Curve Discrete Logarithm Problem (ECDLP) for security [31,33].

Algorithm 4 Map Encoded Integers to Elliptic Curve Points

Input: Set of encoded integers $Q = \{q_1, q_2, \dots, q_t\}$, base point G on elliptic curve

Output: Set of elliptic curve points $P = \{p_1, p_2, \dots, p_t\}$

```

1:  $P \leftarrow \emptyset$ 
2: for all  $q_i \in Q$  do
3:    $p_i \leftarrow q_i \cdot G$  ▷ Scalar multiplication on elliptic curve
4:    $P \leftarrow P \cup \{p_i\}$ 
5: end for
6: return  $P$ 

```

This step ensures non-invertibility under normal assumptions of ECC, adding a cryptographic layer to the biometric template. The scalar multiplication process is deterministic, collision-resistant, and robust against brute-force attacks. It also ensures the cancelability of the fingerprint template [1,4].

4.4. Enrollment

First, the degree k of the polynomial $F(x)$ is decided. Then $k+1$ random integers are generated, which serve as the secret $S = \{s_1, s_2, \dots, s_k, s_{k+1}\}$ which is used as the coefficients of the Polynomial $F(x)$. Then the secret S , degree k , genuine feature set Q , and number of chaff points n is passed to 1 and the vault is created. The private key s for the user is calculated by concatenating all the values in S :

$$s = s_1 \parallel s_2 \parallel \dots \parallel s_{k+1}$$

From the private key s , we construct the public key for the user by mapping it on the elliptic curve.

$$Pub = s.G$$

This public key is also stored along with the vault, and is used while unlocking the vault.

4.5. Verification

During verification, the query minutiae points are passed through the same quantization and encoding algorithm, and we get query integers set Q' . These integers are then mapped to the same elliptic curve using the same base point G , and query points P' are obtained.

Table 1
Obtained GAR/FAR values on FVC2002 DB1 dataset.

Factor	Degree 5	Degree 6	Degree 7	Degree 8	Degree 9	Degree 10	Degree 11
16	100.0%/0.1%	99.0%/0.0%	99.0%/0.0%	98.0%/0.0%	97.0%/0.0%	97.0%/0.0%	95.0%/0.0%
18	100.0%/0.1%	100.0%/0.0%	100.0%/0.0%	100.0%/0.0%	98.0%/0.0%	98.0%/0.0%	96.0%/0.0%
20	100.0%/0.3%	100.0%/0.1%	100.0%/0.1%	99.0%/0.0%	99.0%/0.0%	99.0%/0.0%	98.0%/0.0%
22	100.0%/0.7%	100.0%/0.3%	100.0%/0.1%	100.0%/0.1%	99.0%/0.0%	99.0%/0.0%	96.0%/0.0%
24	100.0%/1.4%	100.0%/0.5%	100.0%/0.2%	100.0%/0.1%	100.0%/0.0%	100.0%/0.0%	98.0%/0.0%
26	100.0%/2.8%	100.0%/1.2%	100.0%/0.5%	100.0%/0.2%	99.0%/0.1%	99.0%/0.0%	99.0%/0.0%
28	100.0%/4.6%	100.0%/2.4%	100.0%/1.2%	100.0%/0.6%	100.0%/0.2%	100.0%/0.1%	99.0%/0.0%
30	100.0%/7.0%	100.0%/3.8%	100.0%/2.1%	100.0%/1.0%	100.0%/0.5%	100.0%/0.2%	100.0%/0.1%

The x-coordinate values of P' , degree k and vault V is passed to 2, which interpolate polynomial using Lagrange interpolation. The coefficients of the generated polynomial is used to yield the secret s' . This s' is mapped to the curve and Pub' is calculated –

$$Pub' = s'G$$

If this public key is same as the saved public key, the user is verified else rejected.

5. Experiment results

A number of performance indicators, such as Equal Error Rate (EER) and Receiver Operating Characteristic (ROC) curves, are used to thoroughly assess the performance of the suggested method. We used datasets from the Fingerprint Verification Challenge (FVC) 2002 and FVC 2004 to evaluate the system's performance. In particular, one FVC 2004 dataset, DB1, and three FVC 2002 datasets, DB1, DB2, and DB3, were employed. Each dataset contains 800 photos total, consisting of grayscale fingerprint images from 100 participants, each of whom has eight images.

The proposed method incorporates two critical tuning parameters to enhance both robustness and security. The first parameter is the *quantization factor* (f), which is employed to quantize the minutiae descriptor values (such as position and orientation). This quantization helps mitigate *intra-class variations*, natural variations in fingerprint impressions from the same user. The second parameter is the *degree* (k) of the polynomial $F(x)$, which directly determines the minimum number of genuine points required to reconstruct the polynomial and successfully recover the secret s from the vault. By adjusting k , the system can control its tolerance to *inter-class similarities*, accidental similarities between different users, thereby enhancing the overall discriminative power and security of the vault. A higher polynomial degree increases the minimum number of correct matches required, thus reducing the chances of false acceptances.

5.1. Setup 1

In the first experimental configuration, each user's first image was used to create the vault, and their second image was used to unlock the vault. The Genuine Acceptance Rate (GAR) and False Acceptance Rate (FAR) values (in %) for the FVC2002 DB1 dataset are shown in Table 1, which correspond to different values of the two tuning parameters. FAR, which is determined by dividing the number of wrongfully accepted occurrences by the total number of impersonation attempts, it is the probability that the system will mistakenly accept an impostor or unauthorized user. On the other hand, GAR, which is the ratio of really accepted instances to the entire number of genuine attempts, shows the probability that the system will successfully accept a legitimate user. We determined the GAR and FAR for each combination by evaluating eight different values for the quantization factor f and five different values for the polynomial degree k . Similarly, Table 2 displays the GAR and FAR values for the FVC2002 DB2 dataset, Table 3 presents the GAR and FAR values for the FVC2002 DB3 dataset, and Table 4 shows the GAR and FAR values for the FVC2004 DB1 dataset.

Plotting the False Acceptance Rate (FAR) and the False Rejection Rate (FRR) yields the Equal Error Rate (EER). FRR, which is determined by dividing the number of falsely rejected cases by the total number of legitimate tries, is the probability that the system would mistakenly reject a legitimate user. At the intersection of the FRR and FAR curves, the EER is calculated. The system performs so well at this intersection that the likelihood of a false rejection is equal to the likelihood of a false acceptance.

On FVC2002 DB1, the best result in terms of EER is 0.005% at $f = 24$ and $k = 10$. Similarly, we obtained 0.02% EER for FVC2002 DB2 at $f = 20$ and $k = 8$. For FVC2002 DB3, we obtained 5.0% EER at $f = 28$ and $k = 5$. Finally, for FVC2004 DB1, we obtained 2.404% EER at $f = 22$ and $k = 5$.

Fig. 4 illustrates the FAR versus FRR curves for the FVC2002 and FVC2004 datasets. The point of intersection of these curves represents the Equal Error Rate (EER). From Fig. 4(a), it is observed that the EER for FVC2002 DB1 occurs at $k = 10$ for $f = 24$. Similarly, in Fig. 4(b), the EER point for FVC2002 DB2 is found at $k = 8$ for $f = 20$. For FVC2002 DB3, as shown in Fig. 4(c), the EER occurs at $k = 5$ for $f = 28$. Finally, Fig. 4(d) displays that the EER for FVC2004 DB1 occurs at $k = 5$ for $f = 22$.

The ROC curves for FVC2002 DB1, DB2, DB3, and FVC2004 DB1 are shown in Figs. 5(a), 5(b), 5(c), and 5(d), respectively. For different threshold levels, the Genuine Acceptance Rate (GAR) versus the False Positive Rate (FPR) is shown graphically by the ROC curve. Better fingerprint recognition system performance is shown by a higher Area Under the Curve (AUC) score, which shows a greater capacity to accurately identify true matches while reducing false positives. The AUC values for FVC2002 DB1, DB2, and DB3, as well as FVC2004 DB1, are 0.9999, 1.0, 0.9706, and 0.9885, respectively, as illustrated in Fig. 5. When the classifier's AUC score is near 1, it means that it can discriminate between real and fake fingerprint matches with high accuracy.

Table 5 Compares the suggested fingerprint recognition method with a number of current methods using four benchmark datasets (FVC2002 DB1, DB2, DB3 and FVC2004 DB1) with Equal Error Rate (EER) as the evaluation parameter. In biometric systems, EER is a crucial parameter that shows the point at which the false rejection rate and the false acceptance rate are equal. Improved accuracy and dependability are shown by lower EER values.

With the lowest EERs on FVC2002 DB1 (0.00%) and DB2 (0.02%), the suggested solution achieves near-perfect performance and routinely beats the current methodologies. Compared to higher values such as 6.14% [6] (Sandhya and Prasad) and 11.38% [10] (Jignesh et al.), the suggested method records 5.00% on FVC2002 DB3, which is competitive even though it is not the best of both. The EER of 2.40% for FVC2004 DB1 obtained by the suggested method is higher than that of the majority of the compared procedures.

The performance trends reveal that while some existing methods show strong results on specific datasets, they often lack consistency across all databases. In contrast, the proposed method demonstrates reliable and stable performance throughout, highlighting its robustness and general applicability. Its consistently low EER values across multiple benchmark datasets indicate a high capability to minimize both false acceptances and false rejections. This makes it highly suitable for secure and accurate biometric identification systems.

Table 2

Obtained GAR/FAR values on FVC2002 DB2 dataset.

Factor	Degree 5	Degree 6	Degree 7	Degree 8	Degree 9	Degree 10	Degree 11
16	100.0/0.1313	100.0/0.0606	100.0/0	100.0/0	97.0/0	96.0/0	96.0/0
18	100.0/0.3939	100.0/0.1515	99.0/0.0202	99.0/0.0101	99.0/0.0101	99.0/0.0101	98.0/0
20	100.0/1.1515	100.0/0.4444	100.0/0.1313	100.0/0.0404	99.0/0.0101	99.0/0	98.0/0
22	100.0/2.0101	100.0/1.0505	100.0/0.3232	100.0/0.1515	100.0/0.0808	99.0/0.0303	99.0/0.0202
24	100.0/4.4949	100.0/2.1717	100.0/0.9697	100.0/0.3939	100.0/0.202	100.0/0.1111	100.0/0.0808
26	100.0/7.6465	100.0/3.9697	100.0/2.2727	100.0/1.2525	100.0/0.596	100.0/0.2828	100.0/0.1313
28	100.0/11.2222	100.0/6.9091	100.0/4.303	100.0/2.3939	100.0/1.3838	100.0/0.697	100.0/0.3535
30	100.0/15.7677	100.0/9.9899	100.0/5.9394	100.0/3.6263	100.0/2.2222	100.0/1.2828	100.0/0.7677

Table 3

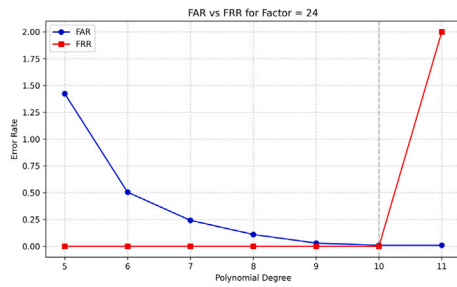
Obtained GAR/FAR values on FVC2002 DB3 dataset.

Factor	Degree 5	Degree 6	Degree 7	Degree 8	Degree 9	Degree 10	Degree 11
16	91.58/0.1411	88.42/0.0868	85.26/0.0434	81.05/0.0326	75.79/0.0217	72.63/0.0217	64.21/0.0217
18	94.74/0.2604	91.58/0.1194	85.26/0.0434	84.21/0.0434	80.0/0.0109	74.74/0	71.58/0
20	94.74/0.6294	92.63/0.2496	91.58/0.1411	87.37/0.0977	83.16/0.0651	82.11/0.0217	78.95/0.0109
22	94.74/1.0635	93.68/0.5534	92.63/0.3581	87.37/0.2062	83.16/0.1194	82.11/0.0868	75.79/0.0326
24	94.74/2.1378	93.68/1.0635	92.63/0.6511	91.58/0.3473	89.47/0.2062	84.21/0.1302	80.0/0.0543
26	94.74/3.5377	94.74/1.9533	93.68/1.096	90.53/0.7379	87.37/0.4124	84.21/0.2604	81.05/0.1628
28	94.74/5.3825	94.74/3.1145	94.74/1.6603	94.74/1.0526	92.63/0.6186	88.42/0.3907	83.16/0.2604
30	96.84/8.1932	95.79/4.6989	94.74/2.7781	93.68/1.8231	89.47/1.1612	87.37/0.7596	86.32/0.4558

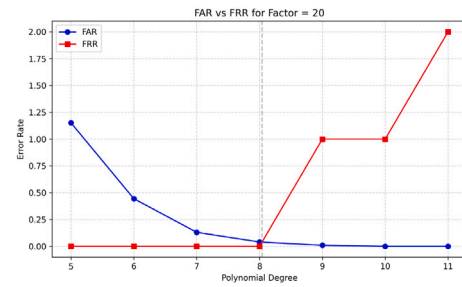
Table 4

Obtained GAR/FAR values on FVC2004 DB1 dataset.

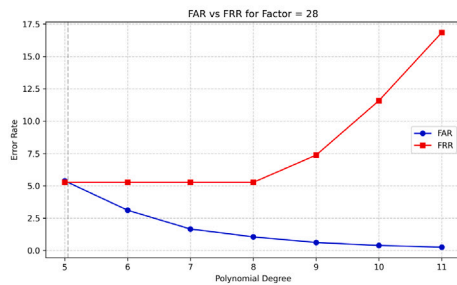
Factor	Degree 5	Degree 6	Degree 7	Degree 8	Degree 9	Degree 10	Degree 11
16	92.0/0.2525	89.0/0.1111	88.0/0.0404	83.0/0.0101	83.0/0	78.0/0	72.0/0
18	95.0/0.8081	94.0/0.3636	91.0/0.1515	88.0/0.0909	86.0/0.0707	85.0/0.0202	79.0/0
20	96.0/1.6061	95.0/0.7273	92.0/0.3333	90.0/0.1414	85.0/0.0808	82.0/0.0202	76.0/0.0101
22	98.0/2.8081	95.0/1.3939	91.0/0.7071	89.0/0.3131	89.0/0.1515	86.0/0.0909	81.0/0.0101
24	95.0/4.4949	93.0/2.4949	93.0/1.2424	90.0/0.5859	88.0/0.3131	85.0/0.1919	80.0/0.101
26	99.0/7.3434	97.0/4.1717	94.0/2.3131	92.0/1.2727	92.0/0.7374	90.0/0.404	84.0/0.2525
28	98.0/9.6263	96.0/5.5253	94.0/3.2222	94.0/1.9394	90.0/1.101	83.0/0.5758	82.0/0.3232
30	99.0/13.2121	99.0/8.2424	97.0/4.7677	92.0/2.9697	92.0/1.8485	89.0/1.1717	85.0/0.7374



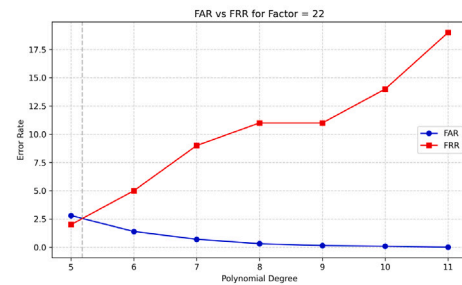
(a) FVC2002 DB1



(b) FVC2002 DB2



(c) FVC2002 DB3



(d) FVC2004 DB1

Fig. 4. FAR/FRR curves of the proposed approach for databases FVC2002 DB1, DB2, DB3, and FVC2004 DB1.

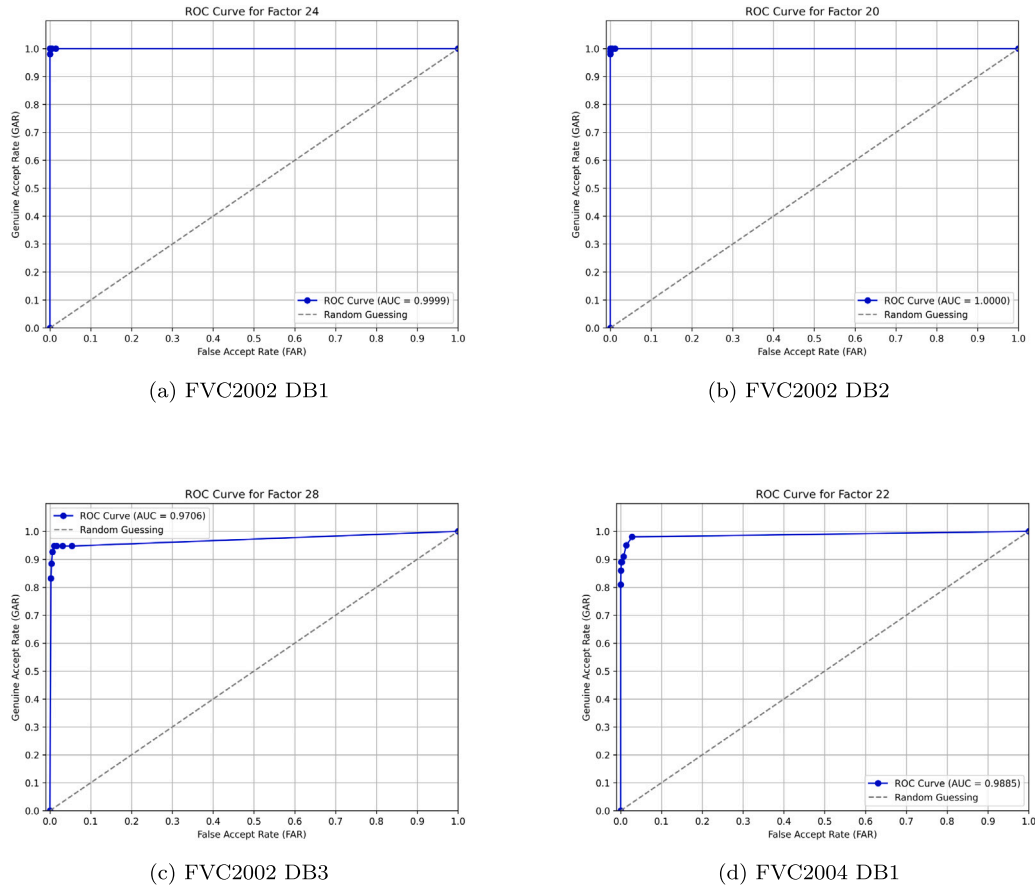


Fig. 5. ROC curves of the proposed approach for databases FVC2002 DB1, DB2, DB3, and FVC2004 DB1.

Table 5

Comparison of the proposed method with existing methods in terms of EER values (in %).

Technique/Author	FVC2002 DB1	FVC2002 DB2	FVC2002 DB3	FVC2004 DB1
Sandhya and Prasad et al. [6]	2.19	1.6	6.14	11.89
Gao et al. [5]	0.24	0.31	2.86	2.44
Vivek et al. [22]	4.00	3.10	–	11.95
M.S.Khan et al. [12]	0.15	0.34	1.28	2.25
M.S.Khan et al. [13]	0.11	0.23	1.02	2.69
S.Song et al. [14]	0.13	0.18	2.50	2.38
Z.Zohu et al. [15]	0.43	2.10	6.60	4.15
Proposed method	0.005	0.02	5.00	2.40

Note: “–” indicates unavailability of data.

5.2. Setup 2

In this experimental setup, we created the vault using the first image of each user, and to unlock it, we tried to unlock the vault using all 8 images of each user one by one. So, there were 800 genuine attempts (for each user there is a vault, so all images are counted as one genuine attempt), and there were 79 200 imposter attempts (as images of one user are imposter attempt for all other users). Table 6 presents the GAR and FAR, expressed in percentages, for the FVC2002 DB1 dataset for various configurations of the two tuning parameters. Specifically, eight values of the quantization factor f and five values of the polynomial degree k were considered. The GAR and FAR values were computed for every possible combination of these parameters to evaluate the system's performance. Similarly, Table 7 displays GAR and FAR values of the FVC2002 DB2 dataset, Table 8 displays GAR and FAR values of the FVC2002 DB3 dataset, and Table 9 displays GAR and FAR values of the FVC2004 DB1 dataset.

In this experimental setup, we achieved the best Equal Error Rate (EER) of 3.54% on FVC2002 DB1 at parameters $f = 30$ and $k = 6$. For

FVC2002 DB2, the minimum EER obtained was 5.24% at $f = 28$ and $k = 6$, while FVC2002 DB3 yielded an EER of 6.73% at $f = 30$ and $k = 5$. Additionally, we achieved an EER of 4.38% on FVC2002 DB4 at $f = 26$ and $k = 7$.

It is important to note that in the FVC2002 databases, impressions 3, 4, 5, and 6 were acquired by instructing users to intentionally vary the placement of their fingers, introducing exaggerated displacement and rotation. As a result, these impressions are of relatively lower quality. Despite this, our system demonstrates strong performance under these challenging conditions, effectively distinguishing between genuine and imposter users. These results highlight the robustness and practicality of our approach, particularly in real-world scenarios where users may not be accustomed to biometric authentication systems.

5.3. Entropy analysis

To quantitatively evaluate the security strength and unpredictability of the generated templates, we adopt the **relative entropy** (also known as Kullback–Leibler Divergence or KLD) as a security metric. This

Table 6

Obtained GAR/FAR values on FVC2002 DB1 dataset.

Factor	Degree 5	Degree 6	Degree 7	Degree 8	Degree 9	Degree 10	Degree 11
16	86.62/0.0404	84.38/0.0088	83.0/0.0038	81.5/0.0013	78.88/0	76.5/0	73.5/0
18	88.5/0.0745	86.62/0.0202	85.25/0.0063	83.38/0	82.0/0	80.0/0	77.62/0
20	89.25/0.1869	88.12/0.0568	86.12/0.0189	84.75/0.0101	83.0/0	81.75/0	79.0/0
22	91.88/0.5707	90.12/0.2033	88.5/0.0758	87.12/0.0278	85.88/0.0051	84.0/0.0013	81.12/0
24	94.25/1.1881	92.25/0.4646	91.0/0.1856	89.62/0.0707	87.5/0.0202	85.12/0.0126	83.5/0.0076
26	94.88/2.3359	93.88/1.0745	92.5/0.4811	91.12/0.2134	89.25/0.0821	87.5/0.0404	85.62/0.0139
28	97.0/4.1086	95.88/2.0379	94.75/0.976	93.38/0.4672	92.25/0.2109	90.25/0.0896	87.0/0.0215
30	97.38/6.3245	96.25/3.3422	94.88/1.7601	93.38/0.8472	92.25/0.4192	90.0/0.1907	87.62/0.0821

Table 7

Obtained GAR/FAR values on FVC2002 DB2 dataset.

Factor	Degree 5	Degree 6	Degree 7	Degree 8	Degree 9	Degree 10	Degree 11
16	90.16/0.0892	88.78/0.0242	87.39/0.0013	85.5/0	83.61/0	81.72/0	79.82/0
18	91.93/0.2318	90.79/0.0764	88.27/0.028	86.38/0.0038	84.99/0.0025	83.23/0.0013	81.08/0
20	91.8/0.7923	91.05/0.3108	88.9/0.1057	88.02/0.0331	87.01/0.0102	85.75/0	83.73/0
22	93.06/1.4725	92.18/0.6738	91.17/0.2611	90.16/0.1006	88.27/0.0395	87.14/0.0178	86.0/0.0089
24	94.2/3.2507	93.06/1.4929	91.93/0.7197	90.79/0.3337	89.91/0.1554	89.03/0.0739	87.64/0.0357
26	95.08/5.9995	93.57/3.215	92.81/1.7272	91.8/0.9451	91.05/0.4624	90.16/0.2267	89.03/0.0955
28	95.59/8.9979	94.83/5.3104	93.95/3.0609	93.57/1.7196	92.31/0.9541	91.42/0.4879	90.67/0.2369
30	95.59/12.6702	94.96/7.7471	94.2/4.6085	93.19/2.7259	92.69/1.568	91.55/0.8904	90.04/0.5146

Table 8

Obtained GAR/FAR values on FVC2002 DB3 dataset.

Factor	Degree 5	Degree 6	Degree 7	Degree 8	Degree 9	Degree 10	Degree 11
16	79.79/0.0902	75.87/0.0389	70.44/0.0171	65.76/0.014	61.84/0.0078	56.71/0.0078	51.58/0.0047
18	83.41/0.2084	79.03/0.0793	73.45/0.0342	69.38/0.0202	64.71/0.0109	60.33/0.0062	56.56/0.0031
20	85.07/0.4696	80.84/0.2021	78.43/0.1057	74.81/0.07	70.14/0.0498	65.76/0.0233	61.84/0.0155
22	87.48/0.9501	85.22/0.4587	82.5/0.2441	78.13/0.1415	72.7/0.0793	68.78/0.0451	64.71/0.0218
24	90.5/1.7851	86.58/0.8863	83.56/0.4525	78.88/0.2597	75.41/0.1353	71.79/0.0746	67.72/0.0451
26	90.95/3.0197	89.74/1.6327	86.88/0.891	83.26/0.5054	77.68/0.2892	73.6/0.1773	69.98/0.1057
28	92.76/4.6571	90.5/2.5781	87.48/1.4181	84.92/0.8179	81.0/0.4976	77.98/0.2939	72.85/0.2021
30	93.67/7.1434	90.8/4.0289	88.84/2.3853	85.52/1.4445	82.2/0.8708	78.43/0.5365	74.66/0.3172

Table 9

Obtained GAR/FAR values on FVC2004 DB1 dataset.

Factor	Degree 5	Degree 6	Degree 7	Degree 8	Degree 9	Degree 10	Degree 11
16	89.18/0.4409	86.92/0.183	83.14/0.0788	78.87/0.0381	75.47/0.0076	71.45/0.0025	67.04/0
18	92.58/1.174	91.07/0.5413	88.3/0.2516	85.53/0.1309	81.51/0.0597	77.11/0.0318	73.71/0.0114
20	94.97/2.3899	92.58/1.1511	90.31/0.5718	88.05/0.2757	85.66/0.141	81.26/0.0534	75.97/0.0267
22	95.6/3.8574	93.71/2.0088	91.32/1.0736	88.55/0.5425	86.79/0.2808	83.77/0.1474	80.0/0.0648
24	95.85/6.5383	94.47/3.7126	92.96/2.0914	90.82/1.2045	88.43/0.6975	86.04/0.3964	82.89/0.2414
26	97.86/10.209	96.73/6.1851	94.84/3.6122	92.7/2.1981	90.94/1.3608	88.18/0.8386	83.14/0.4993
28	97.86/13.6345	96.86/8.6399	94.97/5.2513	93.58/3.2311	90.94/1.9465	87.92/1.1537	85.53/0.6569
30	98.62/18.427	97.74/12.2813	96.98/8.0427	94.97/5.0149	93.21/3.1586	89.56/1.9999	86.54/1.2299

approach is a standard for robustly measuring biometric information, as it accounts for both intra-user and inter-user variability. The relative entropy is given by:

$$KLD = \sum_{x \in X} P(x) \log_2 \left(\frac{P(x)}{Q(x)} \right) \quad (2)$$

where $P(x)$ is the probability distribution of the transformed minutiae features (the x-coordinates of the ECC points) for a legitimate user, and $Q(x)$ is the corresponding feature distribution for the impostor population.

We let $P(x)$ be the feature distribution of the enrolled user, and $Q(x)$ be that of the reference users, i.e., all the other users from the dataset. A higher KLD, measured in bits, indicates a more unique and less predictable template, making it more robust against brute-force or guessing attacks. Table 10 displays the entropy values of the FVC2002 & FVC2004 datasets.

Based on the strong performance in terms of EER and the non-invertible cryptographic transformation, we expect the entropy of our

Table 10

Entropy values of FVC2002 & FVC2004 datasets.

Dataset	Entropy value
FVC2002_DB1	1.215
FVC2002_DB2	1.172
FVC2002_DB3	1.184
FVC2004_DB1	1.218

proposed method to be highly competitive, significantly exceeding traditional biometric systems.

The results in Table 10 demonstrate two crucial properties of our proposed method:

1. **Distinguishability:** The consistently positive KLD values provide the information-theoretic validation that a genuine user's template is statistically distinguishable from the general population of imposters.

Table 11
Comparison of entropy across authentication methods.

Authentication method	Entropy (bits)	Entropy type	Source(s)
Keystroke dynamics	3.48–4.62	Relative (KLD)	Sae-Bae et al. [36]
Face	37.0–55.6	Relative (KLD)	Adler et al. [37]
Iris	278–288	Relative (KLD)	Youmaran et al. [38]
Traditional fingerprint	18.6	Relative (KLD)	Takahashi et al. [39]
Proposed method (Average)	~1.2	Relative (KLD)	This work

2. **Robustness:** The high stability of the entropy values across four different and challenging datasets i.e. FVC 2002 DB1, DB2, DB3 and FVC 2004 DB1 (with an average of ~1.2 bits) underscores the robustness of our feature extraction and protection scheme. The security level does not degrade significantly under different conditions.

5.3.1. Comparative analysis of entropy

Table 11 presents a comparative analysis of the average entropy of the proposed method against commonly used authentication factors, based on values reported in established literature. The table clearly distinguishes between Shannon Entropy and Relative Entropy to ensure a rigorous and scientifically valid comparison.

5.3.2. Interpretation of results

The comparative analysis in Table 11 positions our work effectively. The most important conclusion is not a direct numerical comparison, but the relationship between the theoretical entropy and the practical performance. The relative entropy of ~1.2 bits is the quantitative, theoretical foundation that proves our templates contain unique, identifying information.

The ultimate validation of our method lies in how efficiently this information is used. Our system's ability to leverage these ~1.2 bits of information to achieve exceptionally low Equal Error Rates (EERs), such as 0.005% on FVC2002 DB1 and 0.02% on FVC2002 DB2 demonstrates the high efficacy of our combined ECC and Fuzzy Vault scheme. This synergy between a provable information-theoretic security basis and superior empirical accuracy confirms the overall strength and effectiveness of our proposed method.

6. Security analysis

A thorough security assessment of the suggested approach is provided in this section, with special attention paid to three crucial characteristics: revocability, non-invertibility, and unlinkability.

6.1. Threat model

We use a formal threat model to give a systematic evaluation of our suggested system's security. This model uses the framework developed by Ratha et al. [40], which identifies eight critical points of attack in a biometric system, to define the adversary's capabilities and analyze potential vulnerabilities (see Fig. 6).

Adversary capabilities

The following capabilities are assumed for a strong and realistic adversary:

- The attacker can read, replay, alter, and intercept any data sent over the communication channel between the client (user's device) and the server (database). It is a common *man-in-the-middle* assumption.
- The attacker can access the server-side database in full read mode. Thus, they are able to take all of the users' fuzzy vaults (V) and associated public keys (Pub).

- The attacker is aware of every detail of the proposed system's architecture, including the fuzzy vault construction algorithm, quantization method, feature extraction procedure, and ECC parameters. This follows **Kerckhoffs's Principle**, where security relies on the secrecy of the key/biometric rather than obscurity of the algorithm.

Adversary limitations

The adversary is assumed to be unable to:

- Physically access or compromise the client device. Therefore, they cannot steal the secret key (s) from the client's memory or install a Trojan horse on the feature extraction module.
- Obtain the user's actual biometric trait (e.g., fingerprint) to conduct a spoofing attack. While spoofing is a real threat, defenses such as liveness detection are considered outside the scope of this template protection scheme.

We assess our system's resilience against the eight attack points identified by Ratha et al. [40] using this threat model:

1. **Sensor Attack (Spoofing):** An attacker presents a fake fingerprint to the sensor. Our scheme, which focuses on template protection, does not inherently defend against this. This vulnerability is typically addressed by liveness detection mechanisms at the hardware or sensor level. In our proposed method, the attacks performed on the data at sensor point are not protected as they are hardware oriented.
2. **Channel between Sensor and Feature Extractor:** An attacker might try to replay raw fingerprint data that has already been intercepted. According to the constraints of our model, this attack takes place on the client device, which we presume to be secure.
3. **Feature Extractor Attack (Trojan Horse):** An attacker could compromise the feature extraction module to output a predetermined set of minutiae. This is also a client-side attack and is outside the scope of our threat model. We do not consider this attack in our proposed method as we assume that the client software is secure.
4. **Channel between Feature Extractor and Vault Creation:** Before the vault is built, our system performs feature extraction, quantization, ECC mapping, and polynomial projection on the client device. There is no transmission of raw or unprotected feature data. Consequently, because the sensitive operations are localized, this channel is not vulnerable in the traditional sense.
5. **Matcher Attack (Fuzzy Vault Unlocking):** The vault unlocking mechanism is the "matcher" in our system. In order to track the changes in matching score, an attacker could try a **hill-climbing attack** by repeatedly altering a synthetic template. Due to the fuzzy vault scheme's lack of a similarity score, our system is resistant to this. Unlocking has a binary outcome: either the polynomial is successfully reconstructed (Success) or it is not (Failure). The attacker is unable to cleverly evolve a template due to the absence of immediate feedback.
6. **Stored Template Attack (Database Breach):** If an attacker breaches the database, they acquire the fuzzy vault V and the public key Pub . The security relies on two principles:

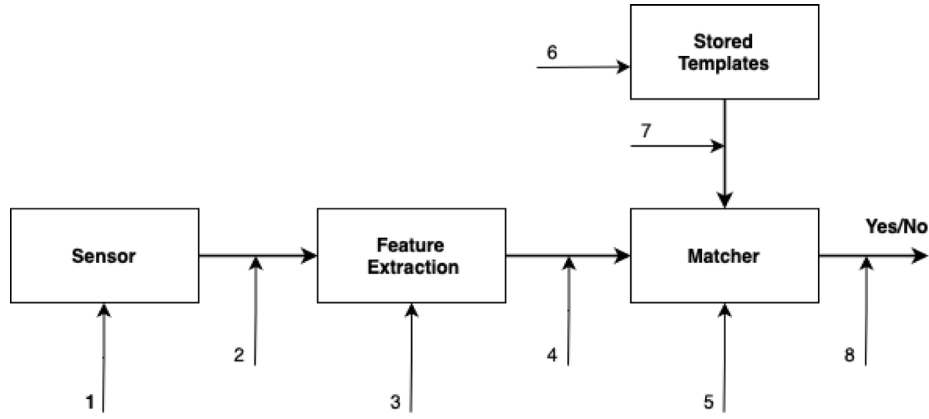


Fig. 6. The eight potential attack points in a generic biometric system, as proposed by Rath et al. [40].

- **Fuzzy Vault Security:** The vault contains genuine points hidden among a large number of chaff points. As demonstrated in our **Brute-force Attack analysis** (Section 6.5), distinguishing the genuine points to reconstruct the polynomial is computationally infeasible without a valid query template.
- **ECC Security:** The stored public key *Pub* does not reveal the secret polynomial coefficients *s* due to the hardness of the Elliptic Curve Discrete Logarithm Problem (ECDLP), as detailed in our *Non-invertibility analysis* (Section 6.3).

7. **Channel between Database and Matcher:** The transmission of the vault from the database to the unlocking module could be intercepted by an attacker. A replayed vault is ineffective on its own since the unlocking process needs a query template from a real user. This offers no new benefit because it is already presumed that the entire vault has been compromised (per attacker's capability #2).
8. **Final Decision Attack:** The final (Accept/Reject) decision is intercepted and changed by an attacker. Standard secure communication protocols like TLS/SSL, which should be implemented at the system level and are not dependent on the biometric template protection scheme itself, address this vulnerability.

In conclusion, our suggested approach offers strong defense against the two most important risks for a template protection system: attacks on the matching module (Point 5) and compromise of the stored template (Point 6).

6.2. Unlinkability

The feature that guarantees a biometric template cannot be linked to a person's true identity or correlated with templates kept in other applications or systems is known as unlinkability in biometric systems. This feature ensures that even if a template is compromised in one situation, it cannot be used to connect to biometric records or betray the identity of the user in another. In order to protect user privacy and prevent unwanted tracking, profiling, or cross-matching between databases, unlinkability is crucial.

The proposed method guarantees that templates created using identical fingerprint biometric information cannot be connected to other systems or applications. For authentication, the templates are stored in a vault and are constructed by concatenating minutiae descriptors after quantization using a factor *f*, followed by mapping onto an elliptic curve with a designated base point *G*. Consequently, under the proposed system, the templates stored in different systems remain unlinkable, as detailed below:

A quantization factor *f* is used to quantize the values of the minutiae descriptors. So for different systems, this quantization factor can be

changed, so the concatenated quantized value will be different even for the same minutiae point $M_i(x_i, y_i, \theta_i)$.

For a given set of genuine minutiae points, let us assume that the quantization factors f_1 and f_2 are used for encoding the minutiae descriptors. The application of these factors leads to two different quantized representations of the same minutiae set. Mathematically, the quantized values for each minutia point can be represented as:

$$(x'_i, y'_i, \theta'_i) = \left(\frac{x_i}{f_1}, \frac{y_i}{f_1}, \frac{\theta_i}{f_1} \right) \quad \text{for } f_1$$

$$(x''_i, y''_i, \theta''_i) = \left(\frac{x_i}{f_2}, \frac{y_i}{f_2}, \frac{\theta_i}{f_2} \right) \quad \text{for } f_2$$

where (x_i, y_i, θ_i) are the original minutiae coordinates and orientation angles. After applying the quantization factors f_1 and f_2 , the resulting quantized minutiae descriptors (x'_i, y'_i, θ'_i) and $(x''_i, y''_i, \theta''_i)$ will differ significantly.

Next, each quantized value is converted into a fixed *b*-bit integer. The quantized values for each minutia are represented as:

$$Q'_i = \text{concatenate}(x'_i, y'_i, \theta'_i) \quad \text{for } f_1$$

$$Q''_i = \text{concatenate}(x''_i, y''_i, \theta''_i) \quad \text{for } f_2$$

where M'_i and M''_i are the concatenated integers for each minutia point after quantization and conversion to *b*-bit values. The concatenation process results in a single *b*-bit integer for each minutiae point.

Since the quantization factors f_1 and f_2 are different, the resulting sets of *b*-bit integers M'_i and M''_i for the same minutiae points will be distinct. Therefore, the two sets of minutiae templates generated using f_1 and f_2 will be completely different, even though they represent the same physical fingerprint.

This ensures that even if the biometric data is exposed across different systems or sessions, it remains unlinkable. The difference in quantization factors causes each set to generate distinct integer representations, which prevents the attacker from correlating the templates or linking them back to the original biometric data, thus maintaining privacy and security.

In addition to quantization factors, the choice of elliptic curve and base point plays a critical role in ensuring unlinkability in biometric template protection. Different elliptic curves and base points result in different cryptographic mappings of the same biometric data, further enhancing security and making it impossible to link templates generated from the same biometric data across systems that use different cryptographic parameters.

Consider a system that uses an elliptic curve *E* and a base point *G* to generate a public key corresponding to a user's biometric data. The biometric features (e.g., minutiae points) are mapped onto the elliptic curve, and the resulting public key is used as a cryptographically

protected template. Let P represent the point on the elliptic curve corresponding to a biometric template, which is computed as:

$$P = Q_i \cdot G$$

where Q_i is the concatenated value associated with the biometric data and G is the base point on the elliptic curve.

If two different elliptic curves E_1 and E_2 with base points G_1 and G_2 are used, even for the same concatenated value Q_i , the resulting mapped points will be different:

$$P_{i1} = Q_i \cdot G_1 \quad \text{on curve } E_1$$

$$P_{i2} = Q_i \cdot G_2 \quad \text{on curve } E_2$$

Since the elliptic curve parameters E_1 , E_2 and base points G_1 , G_2 are different, the resulting points P_{i1} and P_{i2} will be entirely distinct, even though they correspond to the biometric data Q_i . This ensures that templates generated on different systems with different elliptic curves or base points will have completely different representations, making it impossible to link the templates.

We follow the general framework and metrics proposed by Gomez-Barrero et al. [41], to run the experiments for unlinkability analysis.

The process entails examining the distributions of non-mated scores (derived from templates of distinct fingerprint images) and mated scores (derived from templates of the same fingerprint image). These distributions are used to compute two important metrics:

1. **Local Linkability**, $D \leftrightarrow (s)$: It evaluates the linkability of the templates in a score-wise basis. $D \leftrightarrow (s) = 0$ should be interpreted as full unlinkability for that particular score s .

2. **Global System Linkability**, D_{sys} : The overall linkability of the system, independent of the score domain of the system at hand, thereby allowing a benchmark among different systems, where a value closer to 0 indicates stronger unlinkability. $D_{sys} = 0$ means that the system is fully unlinkable for the whole score domain.

We used the FVC2002 and FVC2004 datasets for our investigations. Several secure templates were created for every user. After that, we calculated mated and non-mated scores using the procedure described in our approach. Fig. 7 presents the score distributions and the resulting linkability analysis for each database.

The global unlinkability scores obtained from the analysis are $D_{sys} = 0.0829$ for FVC2002 DB1, $D_{sys} = 0.0305$ for FVC2002 DB2, $D_{sys} = 0.0416$ for FVC2002 DB3, and $D_{sys} = 0.0226$ for FVC2004 DB1. It is confirmed that the score distributions for mated and non-mated templates are appropriately separated by these consistently low values across the challenging FVC2002 datasets. This offers compelling quantitative proof that our suggested approach attains a high level of unlinkability, rendering it resistant to privacy and cross-matching attacks.

Furthermore, the differences in the cryptographic transformations due to different elliptic curves or base points provide an additional layer of security. The attacker, even if they gain access to multiple templates from different systems, will not be able to cross-match these templates or reconstruct the original biometric data, as the mapping of the data onto the elliptic curve is fundamentally altered by the choice of parameters.

6.3. Non-invertibility

In cancelable biometrics, non-invertibility means that reconstructing a person's original biometric information from a stored template or its encrypted representation is computationally infeasible. This feature guarantees that no significant information about the underlying biometric characteristics will be revealed by the exposed biometric template, even in the case of a data breach. The template must be created and saved in a way that makes it impossible for adversaries to reverse-engineer the original biometric data, even if they have complete access to the stored data, in order to ensure non-invertibility.

In the proposed method, non-invertibility is enforced using Elliptic Curve Cryptography (ECC), specifically with the brainpoolP256r1 curve from the TinyEC library. After feature extraction and binarization, the fingerprint template is cryptographically bound to a point on the elliptic curve using a 256-bit private key. This binding acts as a one-way, irreversible transformation without knowledge of the private key. The brainpoolP256r1 curve offers a high-security level, conforming to standards set by the German Federal Office for Information Security (BSI). A 256-bit key on this curve provides approximately **128 bits of security**, which translates to an estimated 2^{128} operations needed to derive the private key from a public ECC point.

Even under ideal attack conditions, a brute-force attempt to recover the private key used in the ECC-based transformation would require approximately 3.4×10^{38} operations. Assuming an optimistic scenario where an attacker possesses a machine capable of performing 10^{12} operations per second (1 trillion ops/s), it would still take more than 10^{18} years to exhaust the key space—an amount of time that vastly exceeds the age of the universe.

Additionally, the non-existence of sub-exponential time algorithms for solving the Elliptic Curve Discrete Logarithm Problem (ECDLP) on prime fields provides strong theoretical assurance. In contrast to RSA or some symmetric encryption schemes that have known structural weaknesses or partial vulnerabilities, ECC remains robust and secure under all currently known cryptanalytic techniques.

The transformation of binary fingerprint features into elliptic curve points is inherently non-invertible. During the process, the original fingerprint features are irreversibly lost due to binarization and their binding to elliptic curve points using ECC operations. Without knowledge of the private key used in this binding, there is no mathematically feasible method to reverse the transformation and recover the original features. Furthermore, even partial information leakage is effectively prevented, as there exists no deterministic or direct mapping between the protected template and the original biometric input. This ensures strong resistance against reconstruction or inversion attacks. The **hill-climbing attack** is based on score evolution through the use of several reference templates. As a result, the hill-climbing attack cannot be performed on our proposed technique as the attacker lacks the necessary feedback information to construct an original template. In an **Attack via Record Multiplicity**, the attacker attempts to reconstruct the original biometric template by correlating multiple instances of its transformed versions. However, in our proposed method, even with access to several transformed templates, the attacker is unable to recover the original biometric data due to the strong non-invertible using Elliptic Curve Cryptography, specifically with the brainpoolP256r1 curve and the fuzzy vault encoding.

6.4. Revocability

A crucial feature of biometric template protection systems is revocability, which guarantees that biometric data can be successfully replaced or revoked without forcing the user to re-enroll in the event of a security breach or template compromise. This ensures that the security and privacy of users are maintained, even when a template is exposed to an attacker. In the context of fingerprint template protection, revocability ensures that compromised templates cannot be misused and provides users with a mechanism to securely update their biometric data when necessary.

In the proposed method, revocability is achieved through the use of **Elliptic Curve Cryptography (ECC)**, specifically with the brainpoolP256r1 curve. The fingerprint template is bound to a 256-bit ECC private key, and the transformation from fingerprint features to cryptographic bindings ensures both **non-invertibility** and the ability to update the key when necessary. By generating a unique private key per user, the system allows for the revocation of a compromised template by simply replacing the key, without requiring the re-enrollment of the user.

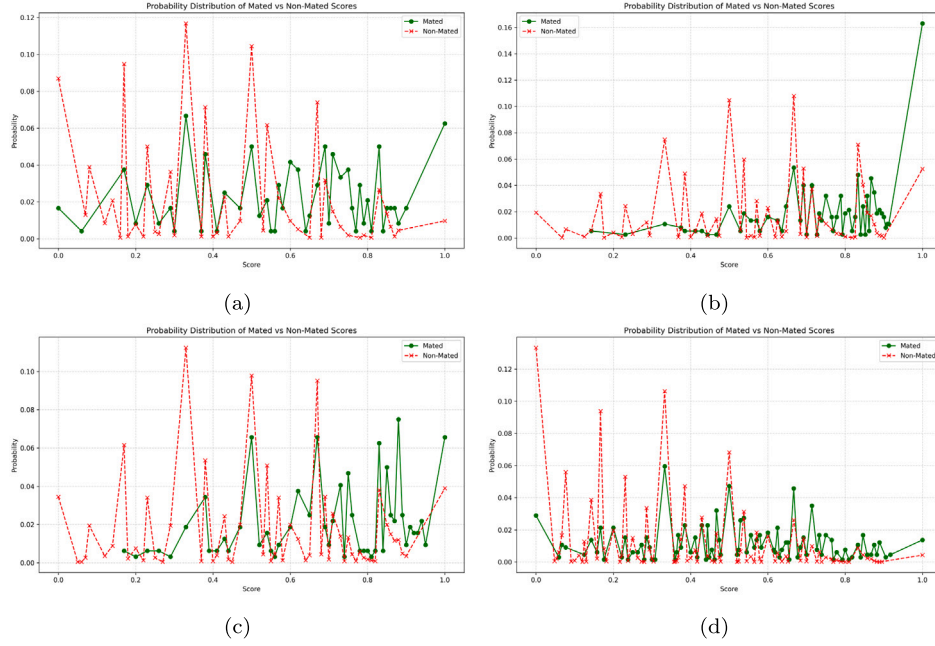


Fig. 7. Unlinkability analysis for (a) FVC2002 DB1, (b) FVC2002 DB2, (c) FVC2002 DB3, and (d) FVC2004 DB1. Each subfigure shows the mated and non-mated score distributions and the local linkability measure $D \leftrightarrow (s)$.

Mechanism of revocation

1. **Key Update:** If a user's template is compromised or if the user requests a template update, the private ECC key associated with the template can be changed. This update renders the old template useless, as the cryptographic binding between the fingerprint features and the key is no longer valid.

2. **Template Regeneration:** In the case of a template update, the system allows for the regeneration of a new template using the latest fingerprint features, which is then bound to a new private key. Even if an attacker manages to obtain an old template, the new key ensures that they cannot exploit it to obtain unauthorized access to the system.

Security and practical considerations

The revocability feature greatly reduces the potential damage caused by a compromised template. Even if an attacker gains access to one template, they cannot reuse it without the corresponding private key, which remains secure. The use of a **256-bit ECC private key** provides significant security, as the computational effort required to break such a key through brute-force or other cryptographic attacks is infeasible with current technology. This ensures that the revocation process is secure and the templates remain protected even in the event of a breach. By enabling template updates through key changes, the system minimizes the risk associated with user re-enrollment, while still maintaining high security and privacy standards.

6.5. Brute-force attack

In the fuzzy vault scheme, a brute-force attack entails an adversary attempting to reconstruct the secret by exhaustively selecting combinations of points from the vault, which contains both genuine and chaff points. The vault is successfully unlocked only if the selected subset includes a sufficient number of genuine points required for polynomial reconstruction. In this context, we analyze the total number of possible point combinations, the probability of a successful decoding attempt, and the computational effort necessary to execute such an attack.

Given the following parameters: The number of genuine points in the vault is $N_{\text{genuine}} = 25$, The number of chaff points in the vault is $N_{\text{chaff}} = 300$, The total number of points in the vault is $N_{\text{total}} =$

$N_{\text{genuine}} + N_{\text{chaff}} = 25 + 300 = 325$, The vault unlocks with any combination of 10 genuine points (out of the 25 available).

The total number of possible combinations an attacker can try is calculated by selecting 10 points from the total of 325 points. This is given by the combination formula $\binom{n}{r}$:

$$\text{Total Combinations} = \binom{325}{10} = \frac{325!}{10!(325-10)!}$$

The value of $\binom{325}{10}$ is approximately:

$$\binom{325}{10} \approx 3.1 \times 10^{18}$$

Thus, the total number of combinations the attacker must evaluate is approximately 3.1×10^{18} .

On the other hand, the number of successful combinations, where the vault is successfully decoded, is the number of ways to choose 10 genuine points from the 25 available genuine points:

$$\text{Successful Combinations} = \binom{25}{10} = \frac{25!}{10!(25-10)!}$$

The value of $\binom{25}{10}$ is approximately:

$$\binom{25}{10} \approx 3,268,760$$

Thus, there are approximately 3.2×10^6 successful combinations that would decode the secret.

Next, the probability of successfully decoding the secret with a random combination of points is the ratio of successful combinations to total combinations:

$$P_{\text{success}} = \frac{\text{Successful Combinations}}{\text{Total Combinations}} = \frac{\binom{25}{10}}{\binom{325}{10}} \approx \frac{3,268,760}{3.1 \times 10^{18}} \approx 1.0 \times 10^{-12}$$

This is quite negligible. Thus it is computationally infeasible for an intruder to unlock the vault by using the brute force attack.

The expected number of combinations that must be evaluated to successfully decode the secret is the inverse of the probability of success:

$$\text{Expected Evaluations} = \frac{1}{P_{\text{success}}} = \frac{1}{\binom{25}{10} / \binom{325}{10}} \approx \frac{3.1 \times 10^{18}}{3,268,760} \approx 1.0 \times 10^{12}$$

Thus, the attacker would need to evaluate approximately 1.0×10^{12} combinations on average.

In conclusion, the total number of possible combinations in this fuzzy vault scheme is approximately 3.1×10^{18} . The probability of successfully decoding the secret with a random combination of points is 1.0×10^{-12} , and the expected number of combinations to evaluate is approximately 1.0×10^{12} .

6.6. Cross-matching attack

A cross-matching attack occurs when an attacker tries to leverage multiple biometric templates stored in different systems for the same user and attempts to reconstruct the original biometric data. The attacker gains access to templates stored in different vaults, such as in a healthcare system or a bank, and attempts to cross-reference the templates to extract the genuine biometric data.

In the proposed method, each system may use different parameters such as elliptic curves, quantization factors, and base points for template generation and encryption. These parameters directly affect the transformation of biometric data into cryptographically protected templates. For instance, varying elliptic curves or base points ensure that even identical biometric data, when processed in different systems, results in different cryptographic representations. Similarly, by using different quantization factors, the concatenated minutiae descriptor values will be different for different systems, and after elliptic curve mapping, a completely different set of points will be generated for the same biometric data.

As a result, even if an attacker acquires multiple templates from different systems, the use of distinct parameters makes it infeasible to cross-match these templates and guess the genuine points or reconstruct the original biometric data. The differences in the cryptographic transformations ensure that the templates remain secure, protecting the genuine points and preventing the attacker from successfully performing a cross-matching attack. Consequently, each system generates a unique and secure template that cannot be used in combination with others to compromise the user's biometric data.

7. Conclusion and future work

By combining the fuzzy vault method and ECC-based mapping, the proposed solution creates a strong foundation for fingerprint-based biometric identification. Important security features, including non-invertibility, revocability, and unlinkability, are guaranteed by this integration. The approach's efficacy is confirmed by experimental assessments on the FVC2002 database, which show low EER and strong discriminability between genuine and fake samples.

In order to create a safe multi-modal recognition system, future research will try to expand the suggested method by combining fingerprint characteristics with other biometric modalities. Combining fingerprint information with other characteristics, such as iris or facial recognition, could significantly increase the authentication process's overall security, resilience, and dependability. This multi-modal approach enables the system to capitalize on the strengths of each modality while mitigating their individual limitations, resulting in a more resilient and versatile authentication framework. Additionally, future research will explore the integration of this biometric system with emerging technologies such as blockchain to enhance data integrity, ensure tamper-proof storage, and provide robust access control mechanisms.

CRedit authorship contribution statement

Praphul Kumar Maurya: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Mulagala Sandhya:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Conceptualization. **I HHNTV Prasad:** Writing – review & editing, Writing – original draft. **Nalaparaju V SuryanarayanaRaju:** Writing – original draft, Software, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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